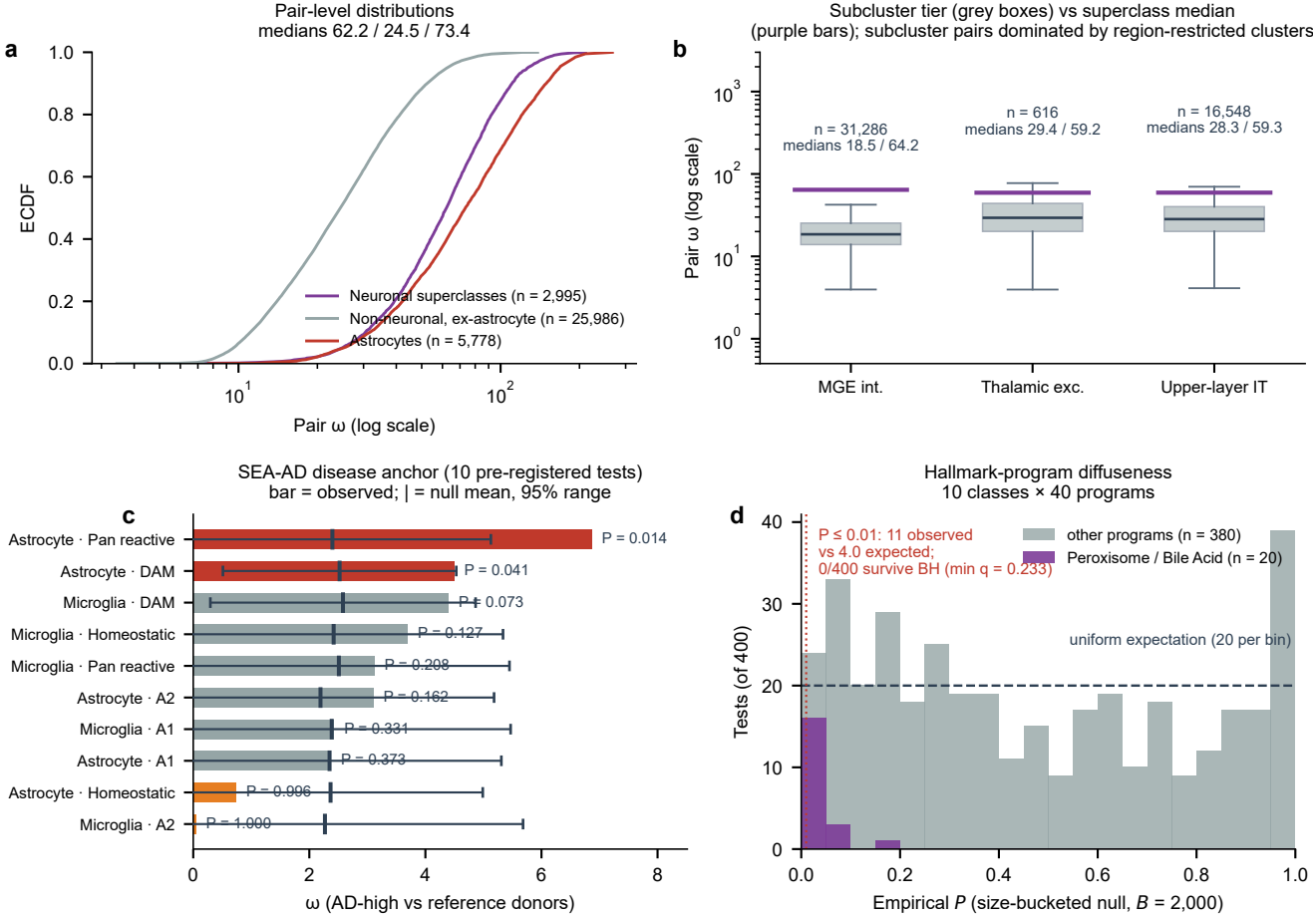

a
Pair-level distributions
medians 62.2 / 24.5 / 73.4
ECDF
Pair ω (log scale)
Neuronal superclasses (n = 2,995)
Non-neuronal, ex-astrocyte (n = 25,986)
Astrocytes (n = 5,778)
b
Subcluster tier (grey boxes) vs superclass median
(purple bars); subcluster pairs dominated by region-restricted clusters
Pair ω (log scale)
n = 31,286
medians 18.5 / 64.2
n = 616
medians 29.4 / 59.2
n = 16,548
medians 28.3 / 59.3
MGE int.
Thalamic exc.
Upper-layer IT
c
SEA-AD disease anchor (10 pre-registered tests)
bar = observed; | = null mean, 95% range
Astrocyte · Pan reactive
Astrocyte · DAM
Microglia · DAM
Microglia · Homeostatic
Microglia · Pan reactive
Astrocyte · A2
Microglia · A1
Astrocyte · A1
Astrocyte · Homeostatic
Microglia · A2
P = 0.014
P = 0.041
P = 0.073
P = 0.127
P = 0.208
P = 0.162
P = 0.331
P = 0.373
P = 0.996
P = 1.000
ω (AD-high vs reference donors)
d
Hallmark-program diffuseness
10 classes × 40 programs
P ≤ 0.01: 11 observed
vs 4.0 expected;
0/400 survive BH (min q = 0.233)
other programs (n = 380)
Peroxisome / Bile Acid (n = 20)
uniform expectation (20 per bin)
Tests (of 400)
Empirical P (size-bucketed null, B = 2,000)